

8.4 Classwork: Rational Functions Challenge — Markscheme

1. $f(x) = \frac{3x-1}{x-2}, \quad x \neq 2.$

(a) Vertical asymptote: $x = 2.$ A1

(b) Horizontal asymptote: $y = 3$ (ratio of leading coefficients $= \frac{3}{1}$). A1

(c) x -intercept: $3x - 1 = 0 \Rightarrow x = \frac{1}{3}$ A1

y -intercept: $f(0) = \frac{-1}{-2} = \frac{1}{2}$ A1

Accept the ordered-pair forms $(\frac{1}{3}, 0)$ and $(0, \frac{1}{2})$.

(d) Let $y = \frac{3x-1}{x-2}$ and swap x and y : M1

$$x(y-2) = 3y-1$$

$$xy - 3y = 2x - 1, \quad y(x-3) = 2x - 1$$

$$f^{-1}(x) = \frac{2x-1}{x-3}$$

(e) Domain of f^{-1} : $x \in \mathbb{R}, x \neq 3.$ A1

FT: award A1 for “ $x \neq$ (the horizontal asymptote value of f)” read from the candidate’s answer in (b).

2. $g(x) = \frac{2-2x}{x+2}, \quad x \neq -2.$

(a) Let $y = g(x)$ and swap x and y : $x = \frac{2-2y}{y+2}$ M1

$$x(y+2) = 2-2y, \quad xy + 2y = 2-2x, \quad y(x+2) = 2-2x$$

$$g^{-1}(x) = \frac{2-2x}{x+2} = g(x)$$

(b) Sketch via GDC (smooth two-branch reciprocal-type curve, upper branch to the left of $x = -2$ heading to $y = -2$, lower branch to the right heading to $y = -2$).

Vertical asymptote: $x = -2$ A1

Horizontal asymptote: $y = -2$ (ratio of leading coefficients $= \frac{-2}{1}$) A1

x -intercept: $2-2x = 0 \Rightarrow x = 1$, so $(1, 0)$ A1

y-intercept: $g(0) = \frac{2}{2} = 1$, so $(0, 1)$ A1

Accept $x = 1$ and $y = 1$ in place of the ordered pairs.

(c) For $h(x) = \frac{2-3x}{cx+d}$:

Vertical asymptote: $x = -\frac{d}{c}$ A1

Horizontal asymptote: $y = -\frac{3}{c}$ A1

(d) A Möbius function $\frac{ax+b}{cx+d}$ satisfies $h^{-1} = h$ iff $a+d=0$.

With $a = -3$, this gives $d = 3$. A1

Accept derivation by swapping $x \leftrightarrow y$ and requiring the resulting expression to equal $h(x)$, leading to $d = -a = 3$.

3. Transform $y = \frac{1}{x}$: vertical stretch $\times 2 \rightarrow$ reflection in x -axis $\rightarrow$ translation 4 right and 3 up. Image $y = p(x)$.

(a) After the three transformations: M1

$$y = \frac{2}{x} \rightarrow y = -\frac{2}{x} \rightarrow y = -\frac{2}{x-4} + 3$$

Combine over a common denominator: M1

$$p(x) = \frac{-2 + 3(x-4)}{x-4} = \frac{3x-14}{x-4} \quad \text{A1} \quad (a=3, b=-14).$$

(b) Asymptotes: $x = 4$ and $y = 3$. A1

Both required for the A1.

(c) x-intercept: $3x - 14 = 0 \Rightarrow x = \frac{14}{3}$ A1

y-intercept: $p(0) = \frac{-14}{-4} = \frac{7}{2}$ A1

Accept $(\frac{14}{3}, 0)$ and $(0, \frac{7}{2})$.

(d) Let $y = \frac{3x-14}{x-4}$ and swap x and y :

$$x(y-4) = 3y-14, \quad y(x-3) = 4x-14 \quad \text{M1}$$

$$p^{-1}(x) = \frac{4x-14}{x-3}, \quad \text{domain } x \in \mathbb{R}, x \neq 3. \quad \text{A1}$$

FT: award A1 for $x \neq$ (the candidate's horizontal-asymptote value from (b)).

4. $f(x) = \frac{1}{x-1}, \quad g(x) = 2x+3.$

(a) $(f \circ g)(x) = f(2x+3) = \frac{1}{(2x+3)-1}$ **M1**

$= \frac{1}{2x+2} = \frac{1}{2(x+1)}$ **A1** ($a=2, h=-1$).

Domain: $x \in \mathbb{R}, x \neq -1.$ **A1**

(b) Re-express as $(f \circ g)(x) = \frac{1}{2} \cdot \frac{1}{x+1}$ **M1**

Two transformations of $y = \frac{1}{x}$:

(i) horizontal translation of 1 unit to the left ($x \mapsto x+1$), **A1**

(ii) vertical stretch with scale factor $\frac{1}{2}$ (equivalently, a vertical compression with scale factor 2). **A1**

Accept the two transformations applied in either order (the order does not matter when one is horizontal and the other vertical).